ICT CORNER

Atomic and Nuclear physics

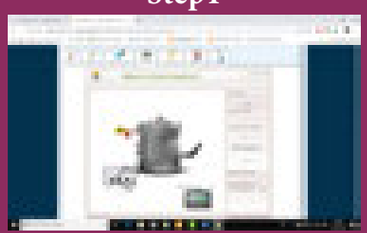
In this activity you will be able to(i) experimentally demonstrate the concept of Millikan's oil drop experiment (ii) find the terminal velocity of the drop and (iii) find the charge on a drop.

Topic: Millikan's oil drop experiment

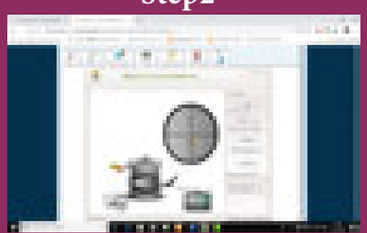
STEPS:

- Open the browser and type"vlab.amrita.edu" in the address bar.
- Click 'Physical Sciences'tab. Thenclick 'Modern Physics Virtual Lab' and then click 'Millikan's oil drop experiment'. Go to "simulator" tab to do the experiment.
- Click on 'START' button. Click on Combo box to choose the oil.
- Click 'START' button of stop watch and notice the time taken t_1 by a drop, to travel distance l_1 between any two points. Calculate the terminal velocity $\upsilon_1 = \frac{l_1}{t_1}$
- Click 'Voltage On' to suspend the same oil drop in air, which is the balancing voltage V.
- Click the 'X Ray ON' button and notice the time taken t_2 by same drop to travel distance l_2 between any two points. Calculate the terminal velocity $\upsilon_2 = \frac{l_2}{t_2}$
- Charge of drop is calculated using the equation $q = \frac{6\pi\eta r(\upsilon_1 + \upsilon_2)d}{V}$. r-radius of oil drop (can be measure using Microscope), η-viscosity of air (1.81×10^{-5} $kgm^{-1}s^{-1}$), d is the distance between the plates.

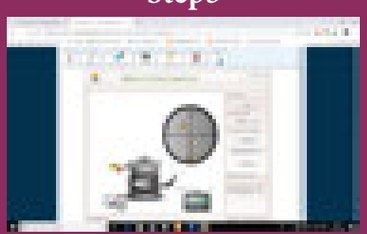
Step1



Step2



Step3



Step4



Note:

1. One time sign up is needed to do simulation. Then login using that username and password.
2. Install flash player in your browserif it is not there.

URL:

http://vlab.amrita.edu/index.php?sub=1&brch=195&sim=357&cnt=4

* Pictures are indicative only.

* If browser requires, allow **Flash Player** or **Java Script** to load the page.

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